Validating Causality

50

po

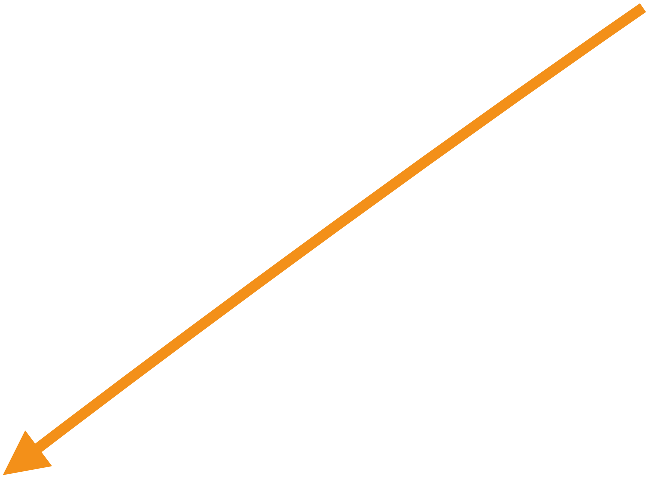
po

po

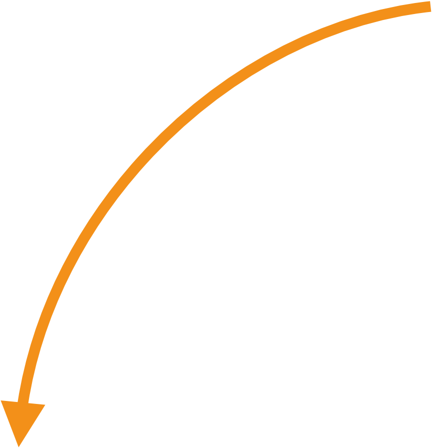
po

po

po



CO_{acc}



COlog

SA+

S A –

R A +

R A

S B +

S B _

C A +

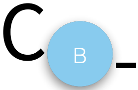
C A –

R B +

R

B

C B +


C
B

```
update(src, +100)
```

```
update(src, -100)
```

```
def update(acc: cown[Account], amount: int) {
  when(acc) { A
    acc.balance += amount
    val id = acc.id()
    when(log) { B log.record(id, amount) }
} }
```

We are given the program events

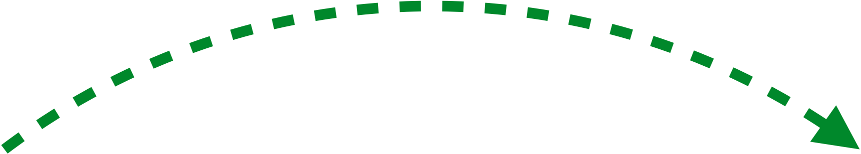
We are given the program and cown orders

We derive the run orders

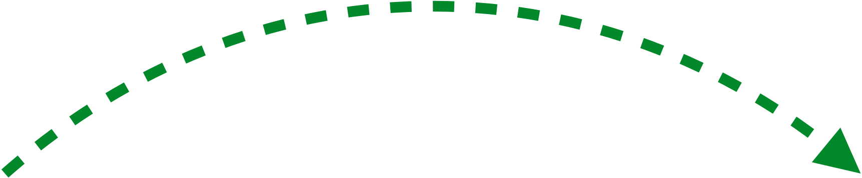
We test for acyclicity

r

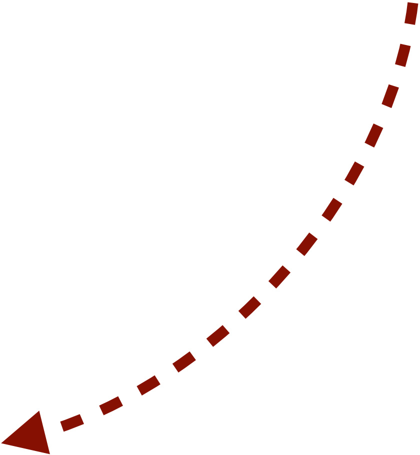
r



r



r



hb

We derive the happens before orders

Validating Causality

```
def update(acc: cown[Account], amount: int) {
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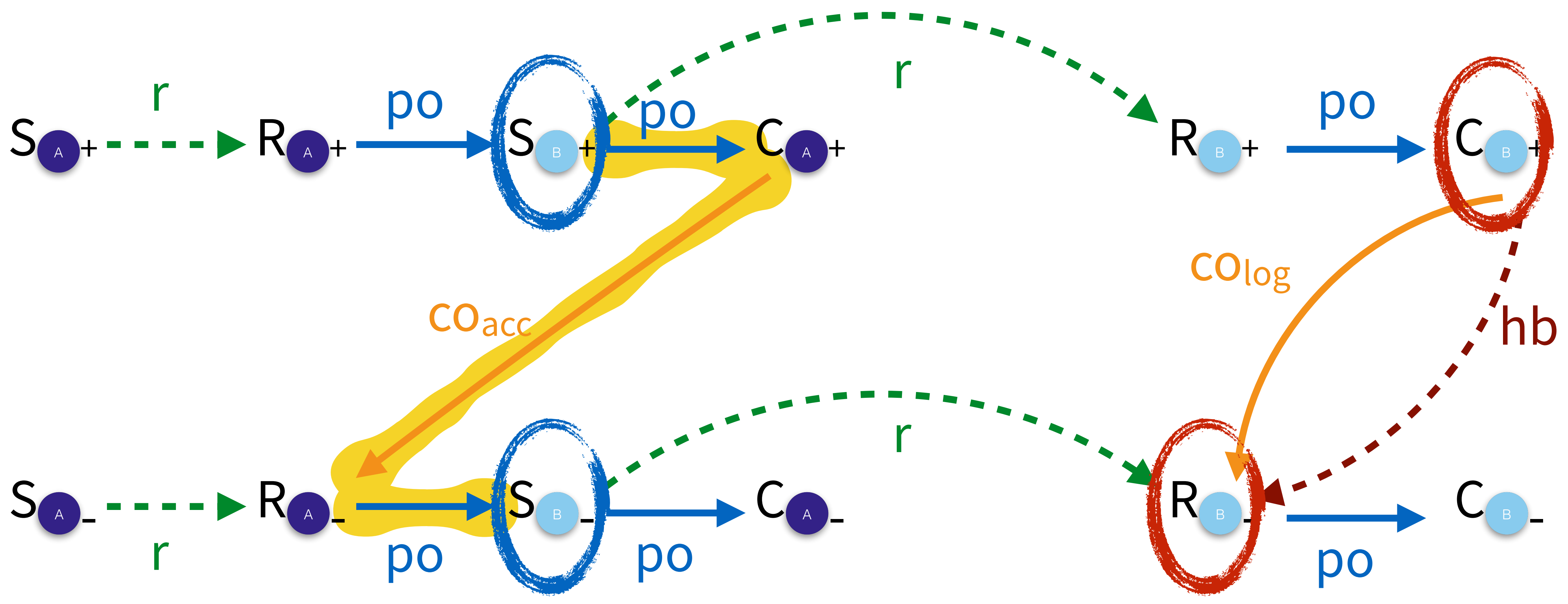
} }
```

```
update(src, +100)
```

```
update(src, -100)
```



We are given the program events

We are given the program and cown orders

We derive the run orders

We derive the happens before orders

We test for acyclicity ✓

Validating Causality

```
def update(acc: cown[Account], amount: int) {
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} }
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